

Cell Datasheet - VBF-T5R2-DSH-01

Case t5_r2 finalist f2_h70_C - NMC811/graphite (Chen2020 base) - simulation-verified only.

Electrical & energy performance

Item	Value	Condition / caliber
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Rated capacity	7.171 Ah	DFN 1C CC discharge 4.2 -> 2.5 V, 25 C (r7)
Nominal capacity parameter	6.5 Ah	parameter-set design value
Nominal voltage	3.78 V	midpoint of 1C DFN discharge
Energy	25.375 Wh	integral V*I over 1C discharge
Energy density (contract)	666.3 Wh/kg	calc-energy caliber: stack only, electrolyte/casing excluded
Energy density, volumetric	1020.5 Wh/L	stack volume 0.025 L
Energy density + electrolyte	~0.7 Wh/kg	BOM-caliber incl. 12.4 g electrolyte fill (1.2 g/cm3 LITERATURE ESTIMATE)
Specific power	42,997 W/kg	contract mass caliber
DC resistance	2.538 mohm	mid-SOC slope estimate (energy tool)
Mass	38.08 g (50.5 g w/ electrolyte)	stack / BOM calibration
Stack dimensions	242.12 um x 0.1027 m2	98.28 um pos / 121.84 um neg / 8 um sep / 8 um Al / 6 um Cu

Operating envelope

Item	Value
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Voltage window	2.5 - 4.2 V (parameter set)
Temperature, ambient	25 - 45 C tested (4C run at 45 C)
Max cell temperature at 4C charge	57.83 C = 330.984 K (ambient 45 C, h = 70 W/m2/K) - passes 60 C limit with 2.2 K margin
Temp at 1C discharge	27.27 C
Charge, standard	1C CC-CV to 4.2 V (recommended taper after cut-off)
Charge, fast	4C CC protocol-verified: no lithium plating (min anode potential +0.0506 V); CC-only acceptance 0.511 Ah before 4.2 V
Discharge, max rated	1C (design basis; higher rates not part of the acceptance criteria)
Cycle life	Not simulated as a rated criterion (requires aging model, artifact-flagged)
Storage	n/a (no calendar-aging criterion)

Cycle-life honesty: this datasheet does NOT fall back on the generic "Not simulated" placeholder, because the Chen2020 base does carry an SEI aging model. An informational aging_1C_100cyc run (SPMe, isothermal 298.15 K, SEI ec-reaction-limited) was executed; the runner's per-cycle capacity series is inconsistent with the contractual 1C capacity (protocol-definition artifact), so only the raw SEI-trend signal is reported: end-SEI 520.3 nm after 100 cycles. No cycle-life rating is derived.

Honesty notes

- All values tool-sourced from r7 finalist simulations (DFN); no post-editing of outputs.
- Electrolyte property values are labeled literature ESTIMATES (endorse skipped, real_compute=false).
- This is a cell-level simulation datasheet, not a certified product specification.